1.5.4 Difference of Two Sets

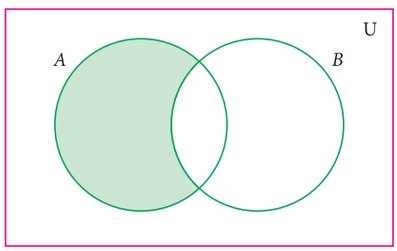
Let A and B be two sets, the difference of sets A and B is the set of all elements which are in A, but not in B. It is denoted by $A-B$ or A\B and read as A difference B.

In symbol, $A-B = \{ x : x \in A \text{ and } x \notin B\}$

$B-A = \{ y : y \in B \text{ and } y \notin A\}$.

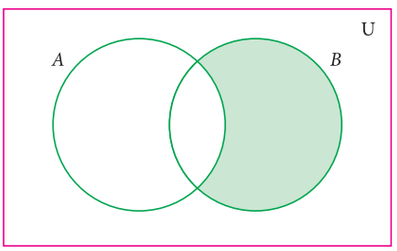
E41LTJ

Venn diagram for set difference



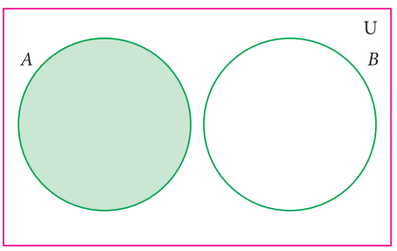
$A-B$

Fig. 1.13



$B-A$

Fig. 1.14



$A-B$

Fig. 1.15

Example 1.14

If $A=\{-3, -2, 1, 4\}$ and $B=\{0, 1, 2, 4\}$, find (i) $A-B$ (ii) $B-A$.

Solution

$A-B = \{-3, -2, 1, 4\} - \{0, 1, 2, 4\} = \{-3, -2\}$

$B-A = \{0, 1, 2, 4\} - \{-3, -2, 1, 4\} = \{0, 2\}$

Note

- $A' = U - A$
- $A-B = A \cap B'$
- $A-A = \varnothing$
- $A-\varnothing = A$
- $A-B = B-A \Leftrightarrow A=B$
- $A-B = A$ and $B-A = B$ if $A \cap B = \varnothing$

1.5.5 Symmetric Difference of Sets

The symmetric difference of two sets A and B is the set $(A-B)\cup(B-A)$. It is denoted by $A \Delta B$.

$A \Delta B = \{ x : x \in A-B \text{ or } x \in B-A\}$

Example 1.15

If $A = \{6, 7, 8, 9\}$ and $B=\{8, 10, 12\}$, find $A \Delta B$.

Solution

$A-B = \{6, 7, 9\}$

$B-A = \{10, 12\}$

$A \Delta B = (A-B)\cup(B-A) = \{6, 7, 9\}\cup\{10,12\}$

$A \Delta B = \{6, 7, 9, 10, 12\}$.

Thinking Corner

What is $(A - B) \cap (B - A)$?

1-SET LANGUAGE.indd 16 26-12-2019 13:32:04